

Program A: Left Rotation of an Array

Question

Given an array of integers, write an algorithm and a C program to left-rotate the array by a specified number of elements.

Algorithm

1. Read the size **n** and the array elements.
2. Read the number of left-rotation positions **d**.
3. Calculate **d = d % n** to ignore complete rotations of the array.
4. Copy the first **d** elements into a temporary array.
5. Shift the remaining **n - d** elements by **d** positions to the left.
6. Copy the saved elements to the last **d** positions of the original array.
7. Display the rotated array.

C Code

```
#include <stdio.h>

#define MAX_SIZE 100

int main(void)
{
    int array[MAX_SIZE], temp[MAX_SIZE];
    int n, d, i;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    if (n <= 0 || n > MAX_SIZE) {
        printf("Invalid array size.\n");
        return 1;
    }

    printf("Enter %d array elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &array[i]);

    printf("Enter positions to left-rotate: ");
    scanf("%d", &d);

    if (d < 0) {
        printf("Number of rotations cannot be negative.\n");
        return 1;
    }

    d = d % n;

    for (i = 0; i < d; i++)
        temp[i] = array[i];

    for (i = 0; i < n - d; i++)
        array[i] = array[i + d];

    for (i = 0; i < d; i++)
        array[n - d + i] = temp[i];

    printf("Array after left rotation:\n");
    for (i = 0; i < n; i++)
        printf("%d ", array[i]);

    printf("\n");
    return 0;
}
```